

# Yuan Si

 [samsi0322.github.io](https://samsi0322.github.io)  [github.com/SamSi0322](https://github.com/SamSi0322)  [Google Scholar](https://scholar.google.com/citations?user=y3si@uwaterloo.ca)  [y3si@uwaterloo.ca](mailto:y3si@uwaterloo.ca)  +1(437)777-5619

## RESEARCH INTERESTS

AI for software engineering (AI4SE), LLM-based program repair, multimodal debugging, program analysis & verification for block-based programs, AI for computing education; analytic number theory, probability theory.

## PUBLICATIONS

- **Yuan Si**, Daming Li, Hanyuan Shi, Jialu Zhang. *VisionScratch: LLM-Based Automated Feedback Generation using Code-Produced Videos for Scratch Programs*. Proceedings of the ACM on Software Engineering, Vol. 3 (FSE 2026), pp. 3534–3557. 
- **Yuan Si**, Simeng Han, Daming Li, Hanyuan Shi, Jialu Zhang. *ScratchEval: A Multimodal Evaluation Framework for LLMs in Block-Based Programming*. , ACM International Symposium on Software Testing and Analysis (ISSTA) 2026 (accepted).
- **Yuan Si**, Jialu Zhang. *Fixed-Set Robustness in Programming by Example: Example Corruption and Semantic Partition Recovery*. , 2026.
- **Yuan Si**, Jialu Zhang. *Checked Program Recovery from Execution Video: A Sound Oracle for Untrusted Generators*. , 2026.
- **Yuan Si**, Jialu Zhang. *SchedCheck: Schedule-Robustness Analysis for Event-Driven Block Programs*. , 2026.
- **Yuan Si**, Jialu Zhang. *Certificate-Carrying Transformation of Event-Driven Block Programs*. , 2026.
- **Yuan Si**, Jialu Zhang. *ScratchLens: Lens-Parametric Behavioral Equivalence for Scratch Programs*. , 2026.
- **Yuan Si**, Ming Wang, Daming Li, Hanyuan Shi, Jialu Zhang. *EcoScratch: Cost-Effective Multimodal Repair for Scratch Using Execution Feedback*. , 2026.
- **Yuan Si**, Kyle Qi, Daming Li, Hanyuan Shi, Jialu Zhang. *Stitch: Step-by-step LLM Guided Tutoring for Scratch*. , 2025.
- **Yuan Si**. *A Microcanonical Phase Transition for the Collatz Affine Random Model* (2026). Submitted to *Forum of Mathematics, Sigma*. 
- **Yuan Si**. *Mixed Parity, Diagonal Denominator, and the Pell-Chord Genus-Five Obstruction for the Four-Corner Rational Distance Problem* (2026). Submitted to *J. Number Theory*.  
- **Yuan Si**. *Elliptic Decomposition of the Pell-Chord Genus-Five Obstruction for the Four-Corner Rational Distance Problem* (2026). 
- **Yuan Si**. *An Elementary Projection Obstruction for Rational Distances to Regular Polygons* (2026). 
- **Yuan Si**. *A Gentle Introduction to Optimization* (2024). Adopted as required reading in 3 university courses. 

## EXPERIENCE

- **University Researcher - AI in Multidimensional Input** Remote, Waterloo, Canada  
*UWaterloo, ECE Department* July 2025 – present
  - Led 9 first-authored papers on LLM-driven debugging, tutoring, evaluation, and formal verification for Scratch (VisionScratch published at FSE 2026, ScratchEval accepted at ISSTA 2026).
  - Built multimodal repair systems fusing gameplay video, execution feedback, and project JSON to localize bugs and synthesize fixes via LLM-guided loops.
  - Developed an interactive tutoring system (Stitch) and a 100-project executable benchmark (ScratchEval) for evaluating LLM repair quality on block-based code.
- **University Researcher - Game Theory & Graph Theory** On-site, Waterloo, Canada  
*University of Waterloo, Combinatorics & Optimization Department* Dec. 2024 – March 2025
  - Derived Nash equilibria for multiplayer RPS via LP ; modeled Tesla charging placement via independent dominating sets ; analyzed Public Goods Games .
- **Microsoft Researcher - AI Trends** Remote, Vancouver, Canada  
*Microsoft - Microsoft Azure & AI, Research* May 2024 – Aug. 2024
  - Investigated AI scribe technologies for healthcare and improved an insurance chatbot via systematic A/B analysis of user interaction data. 

## EDUCATION

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|------------------------------------------------------------------------------------------|--------------------------------|
| <b>University of Waterloo</b>                                                            | Waterloo, Canada               |
| <i>HBM, Math / Data Science &amp; Combinatorics Optimization &amp; Pure Math (co-op)</i> | <i>Sept. 2023 – April 2028</i> |

## PROJECTS

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- **ResearchOS** : Local-first research decision and execution system; turns a brief into runnable experiments, evidence-checked claims, and a packaged manuscript draft. FastAPI + React, provider-agnostic LLM adapters (OpenAI / Anthropic / mock), dual-agent code worker, HITL approval gates, encrypted secret store. AGPL-3.0.
- **DevToolkit** : 43 zero-dependency single-file Python CLI tools across 9 categories (web, data, process, MCP, security, scaffolding).

## SKILLS

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- **Research**: Experiment design, human evaluation, ablation studies, benchmark construction, statistical analysis
- **AI/LLM**: Multimodal repair pipelines, LLM evaluation, prompt engineering, MCP tooling
- **Languages & Tools**: Python, JavaScript/TypeScript, C/C++, Java, Git, Linux, L<sup>A</sup>T<sub>E</sub>X, Magma, SageMath